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Comparison between dual-energy x-ray absorptiometry and bioelectrical impedance for body composition measurements in adults with chronic kidney disease: A cross-sectional, longitudinal, multi-treatment analysis

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ABSTRACT

Objective: The aim of this study was to evaluate the agreement between bioelectrical impedance spectroscopy (BIS) and dual-energy x-ray absorptiometry (DXA) for assessment of body composition in patients with chronic kidney disease (CKD).

Methods: We performed cross-sectional and prospective analyses by DXA and BIS in whole body (BISWB) and segmental (BISSEG) protocols in CKD non-dialysis–dependent (n = 81), hemodialysis (n = 83), peritoneal dialysis (n = 24), and renal transplantation (n = 80) patients. Intraclass correlation coefficient (ICC) and Bland–Altman plots were evaluated. Linear regression analysis was performed for bias assessment and development of equations. Receiver operating characteristics curve was constructed for diagnosis of inadequate error tolerance (DXA − BIS >±2kg).

Results: The agreement with DXA was greater for BISWB than BISSEG; for fat mass (FM; ICC men = 0.894; women = 0.931) than fat-free mass (FFM; ICC men = 0.566; women = 0.525), with greater bias for FFM as muscle increases and for FM in body fat extremes. The agreement was lower for body change analysis (ICC FFM men = 0.196; women = 0.495; ICC FM men = 0.465; women = 0.582). The ratio of extra- to intracellular water (ECW/ICW), body mass index, fat mass index, waist circumference, resistance, and reactance interfered in bias between methods. An ECW/ICW cutoff point of ≥0.7250 for inadequate error tolerance was determined. New prediction equations for FFM (r^2 = 0.913) and FM (r^2 = 0.887) presented adequate error tolerance in 55% and 63% compared with 30% and 39% of the original equation, respectively.

Conclusion: For body composition evaluation in patients with CKD, BIS applied using the whole body protocol, in normal hydration patients with CKD is as reliable as DXA; BIS must be used with caution in overhydration patients with ECW/ICW ≥ 0.7250. The newly developed equations are indicated for greater precision.

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Introduction

Population growth and aging, reduced mortality from communicable diseases, and increased burden of non-communicable diseases from 1990 to 2016 have shaped the epidemiology of chronic kidney disease (CKD), which had an increase in incidence and prevalence [1]. Furthermore, CKD ascended from the 18th leading cause of death in 1990 to the 11th in 2016 [2], showing a lack of progress in disease management.

Patients with CKD frequently have anorexia with nutrient deficiency, an altered body composition driven by metabolic disorders from the uremic syndrome [3] that usually leads to increased protein catabolism, with obesity and muscle impairment being common conditions [4–6] associated with worse clinical prognosis [7,8] and high mortality rates. Therefore, with increased mortality and disability rates of patients with CKD in the last quarter of a century [1] and the association of a poor body composition with low survival, routine and longitudinal assessment of body composition could improve clinical outcomes, as body composition is not only a research tool but also a clinical prognostic factor.

Clinically invisible fluid overload and excess adiposity in CKD can mask changes in body composition by increasing body mass index (BMI). Therefore, body composition, used for guiding clinical management, must be correctly assessed to detect clinically relevant changes. To fulfill the task, appropriate, accurate, and easily available instruments are needed.

Dual x-ray absorptiometry (DXA) is a well-documented, accurate method that is accepted as a reference for body composition evaluation [9–11]. However, DXA machines are expensive, nonportable, and their operation requires specialized staff, which make their use unfeasible in routine clinical practice for assessment of body composition.

On the other hand, bioelectrical impedance devices, originally applied for measurements of hydration status, are inexpensive, non-invasive, portable, fast, and easy to apply for evaluations of body composition. Also, there is an increasing interest in their potential for nutritional assessment [10].

Because of the disadvantages of the reference method (DXA), the most adequate alternative method for body composition assessment of the CKD population needs to be well established. Considering the relevance and feasibility of bioelectrical impedance spectroscopy (BIS) for water imbalance and body composition assessments, its agreement with DXA parameters must be known for appropriate use and settings. Recent studies have evaluated the agreement between BIS with DXA in adults across a spectrum of BMI [12–14], in patients with non-dialysis–dependent (NDD) CKD [15], in patients on peritoneal dialysis (PD) [16], and for body fat measurement in patients on hemodialysis (HD) and in NDD patients [17]. However, to the best of our knowledge, there is no study that evaluated the body composition of patients with CKD in the four types of treatment (NDD, HD, PD, and renal transplant [KTx]) using whole body and segmental BIS, in prospective and longitudinal analysis, and comparing the results with the fat-free mass (FFM) and fat mass (FM) data obtained by DXA.

The aims of this study were to evaluate the agreement between BIS and DXA body composition data of patients with CKD in NDD, HD, PD, and KTx therapy, identify the main factors associated with disagreement between methods and establish a cutoff point for adequate agreement, and develop and validate new prediction equations for body composition evaluation.

Methodology

Study population

This was an observational cross-sectional study followed by a prospective analysis. The sample size calculation was performed based on agreement between DXA and BIS, for each CKD treatment subgroup, with $\alpha = 0.05$, $\beta = 0.20$, test power of 80%, μ/σ (difference limits) = 0.4, and δ/σ (limits of agreement) = 3 [18].

Patients with CKD ($\leq$18 to $\leq$60 y of age) from a tertiary hospital were recruited. Inclusion criteria were CKD patients with any number of comorbidities, in stages 3b to 5 for NDD patients; PD patients without peritonitis in the previous 30 d and HD patients (4-h dialysis session, three times per week, through an arteriovenous fistula), both in dialysis for $\geq$3 mo; and KTx patients for $\geq$6 mo with CKD in stages 1 to 3a. Exclusion criteria were presence of amputated limbs or an electronic implant, wheelchair user or inpatient, body weight >140 kg or BMI >40 kg/m^2, acute infections, cancer diagnosis, AIDS, and other conditions could alter body composition.

Residual renal function was estimated using mean urinary creatinine clearances adjusted for 1.73 m^2 obtained from 24-h urine collection performed 1 d to 1 wk before study assessment. Kt/V was estimated by weekly clearance of dialyzed urea adjusted by total body water (TBW) in the week before the study.

The local Ethics Committee approved the study. All participants read and signed the informed consent form before the procedures began. This study was conducted according to the guidelines laid down in the Declaration of Helsinki and all procedures involving human subjects/patients were approved by the Ethics Committee of the Ribeirão Preto Faculty of Medicine Clinics Hospital.

Clinical and biochemical data and self-declared ethnicity were collected from electronic medical records up to 10 d before the study. Anthropometric (weight at the nearest 0.1 kg, height and waist circumference [WC] at the nearest 0.1 cm), and BIS and DXA measurements were performed consecutively on the same day by the same technician after an 8-h fast, empty urinary bladder, drainage of the peritoneal dialysate, and just after the midweek HD session. Measurements were performed with patients wearing a light weight gown, without shoes, and preferably on the right side of the body (except if a fistula was present).

Body composition assessment

DXA (Hologic Discovery A, USA, Bedford, MA) was used to evaluate lean mass (LM), appendicular LM index (ALMI), FFM (sum of LM with bone mineral content), and FM, with segmental evaluation of both arms, both legs, and trunk as well as total masses by the adding of segments [9]. A whole body scan was done after daily calibration of the device by scanning a spine phantom. ALMI was determined by the sum of LM of both arms and legs normalized for squared height, as also was done for LM (LMI), FFM (FFMI), and FM (FMI) indexes.

BIS (BCM, Fresenius Medical Care, Bad Homburg, Germany) was performed in a tetra-polar unilateral whole body wrist-to-ankle protocol [19] (BISWB) after a 10-min adaptation in a supine position. The variables resistance (Re), reactance (Xc), and phase angle (PhA) were provided at 50 kHz frequency; TBW, extracellular water (ECW), and intracellular water (ICW), as well as the ECW/ICW ratio, overhydration (OH), body cell mass (BCM), FFM, FFMI, FM, and FMI were estimated by the equipment's prediction formulas [20,21]. A segmental protocol (BISSEG) was also used [22], providing FFM and FM data of right arm, right leg, and trunk, as well as total FFM and FM calculated by the sum of segments (2 × right arm + 2 × right leg + trunk) and estimated using previously published formulas [22,23], applying weight (5, 16, and 50% of total body weight for arm, leg, and trunk, respectively) [24], height (mean of a duplicate measure) [22], Re, and Xc of each segment.

For PhA classification, reference values of a healthy population of Hispanic ethnicity were applied (PhA <6.22° and <5.14° for men and women, respectively) [25].

Hydration status rating was established by OH index provided by BISWB, based on a physiologic tissue model that involves the individual's normal ECW, normal-hydration adipose tissue and lean tissue, calculated from the difference between measured and normal expected ECW [20,21]. OH and dehydration were considered if OH > +1.1 L and < −1.1L, respectively [26].

WC was measured with a flexible non-stretchable plastic tape at umbilical scar level [27].

Statistical analysis

Data normality was checked using the Shapiro–Wilk test. Data are presented as mean ± SD or number (n) and percentage. To analyze differences among subgroups according to sex and CKD treatment, unpaired *t* test or analysis of variance (ANOVA; quantitative variables) and χ^2 (qualitative variables), adjusted by Bonferroni post-test were applied.

Agreement in FFM and FM measurements between DXA and BIS, for cross-sectional data and for body composiiton changes (prospective measurement − cross-sectional measurement [Δ]) was checked using intraclass correlation coefficient (ICC) and 95% confidence intervals (CIs) with values close to 1 indicating

Table 1
Descriptive analysis of total sample and stratification by CKD subgroup

Variables	Categories	NDD (n = 83)		HD (n = 79)		PD (n = 23)		KTx (n = 81)		Total sample (N = 266)	
		X	SD	X	SD	X	SD	X	SD	X	SD
Age (y)	–	48[a]	10	47[a]	10	40[b]	12	49[a]	8	47	10
Ethnicity*, % (n)	White	73 (61)[a]		71 (56)[a]		74 (17)[a]		74 (60)[a]		73 (194)	
Sex, % (n)	Male	55 (46)[a]		44 (35)[a]		35 (8)[a]		59 (48)[a]		51 (137)	
DM, % (n)	Present	30 (25)[a]		13 (10)[b]		4 (1)[b]		43 (35)[a]		27 (71)	
SAH, % (n)	Present	73 (61)[a]		67 (53)[a]		61 (14)[a]		64 (52)[a]		68 (180)	
DLP, % (n)	Present	23 (19)[a]		15 (12)[a]		9 (2)[a]		27 (22)[a]		21 (55)	
Dialysis or kidney transplant time (mo)	–	NA		74[†]	62	15[†]	17	92	61	NA	
eGFR (mL/min/1.73m^2)	–	18.6[†]	8.52	NA		NA		70.37[†]	18.39	NA	
KT/V	–	NA		1.68[†]	0.58	2.59[†]	0.51	NA		NA	
Cr (mg/dL)	–	3.7[a]	1.30	10.82[b]	3.17	8.61[c]	3.08	1.33[d]	1.68	5.51	4.53
Weight (kg)	–	77[a]	17	66[b]	13	65[b]	15	71[b]	13	71	15
BMI (kg/m^2)	–	28[a]	6	25[b]	5	24[b]	4	27[ab]	4	26	5
WC (cm)	–	102[a]	15	95[bc]	13	93[b]	11	100[ac]	9	99	13

BMI, body mass index; CKD, chronic kidney disease; Cr, creatinine; DLP, dyslipidemia; DM, diabetes mellitus; eGFR, estimated glomerular filtration rate; HD, hemodialysis; KTx, kidney transplant; NA, not applicable; NDD, non-dialysis–dependent; PD, peritoneal dialysis; SAH, systemic arterial hypertension; WC, waist circumference.
Analysis of variance (quantitative variables) and χ^2 (qualitative variables) between CKD subgroups with adjusted P by Bonferroni test, $P \leq 0.008$ (values without a common letter [abcd] differ significantly)
*Ethnicity: white or non-white (black, mulatto, or dark skinned).
[†]Unpaired Student's t test between HD and PD or NDD and KTx groups, $P \leq 0.05$.

high agreement [28]. Agreement on the individual level was evaluated using the Bland–Altman method with limits of agreement (LOAs). The 5% error tolerance between DXA and BIS ($\leq$0.95 BIS/DXA $\leq$ 1.05) was calculated. Linear regression analyses were performed to determine the proportional bias between methods.

The total sample was randomly divided in a ratio of 75% for design and 25% for validation with stratification by ECW/ICW ratio, FMI, and sex. Linear regression analysis was used for development and validation of new prediction equations for FFM and FM.

The total sample was split into tertiles of ECW/ICW ratio and analyses of agreement between DXA and BIS were performed and compared among tertiles. Also, receiver operating characteristics curve (ROC) was applied for ECW/ICW ratio, and inadequate error tolerance (DXA–BIS > ±2 kg) diagnosis was applied and validated for FFM.

Differences between design and validation groups were compared by unpaired t test.

Significance was set at $P \leq 0.05$ except if adjustment for multiple comparisons were necessary.

All statistics were performed with SPSS version 23 (IBM, Armonk, NY, USA).

Results

Characteristics of the study population

The baseline clinical characteristics of the total sample and CKD subgroups are shown in Table 1. The total sample was comprised of 51% men; 31%, 30%, 9%, and 30% were in NDD, HD, PD, and KTx CKD subgroups, respectively. CKD was secondary to the following conditions:

- Systemic arterial hypertension (SAH) in 29%
- Glomerulonephritis in 25%;
- Diabetes mellitus (DM) in 10%;
- Polycystic kidney in 7%;
- Glomerulosclerosis and systemic syndromes in 8%; and
- Other causes and unknown etiology in 20%.

Table 2
Bioelectrical impedance data by BISWB of total sample and stratification by CKD subgroup

Variables	Categories	NDD (n = 83)		HD (n = 79)		PD (n = 23)		KTx (n = 81)		Total sample (N = 266)	
		X	SD	X	SD	X	SD	X	SD	X	SD
Re 50 kHz (ohm)	–	487[a]	103	590[b]	93	544[ab]	96	524[a]	88	534	104
Xc 50 kHz (ohm)	–	51[a]	14	60[b]	14	55[ab]	13	54[a]	8[b]	55	13
PhA (°)	–	5.94[a]	0.97	5.85[a]	1.17	5.80[a]	0.93	5.97[a]	0.83	5.91	0.99
PhA classification[1] % (n)	Low PhA	32 (27)[a]		34 (27)[a]		35 (8)[a]		35 (28)[a]		34 (90)	
BCM (kg)	–	24.18[a]	6.69	19.88[b]	6.62	22.20[ab]	6.22	22.14 [ab]	6.87	22.11	6.96
ECW (L)	–	17.26[a]	3.78	13.84[b]	3.10	15.12[abc]	3.49	15.53[c]	2.99	15.53	3.57
ICW (L)	–	20.60[a]	4.28	17.59[b]	4.05	18.70[ab]	4.33	19.03[ab]	4.16	19.06	4.32
TBW (L)	–	37.86[a]	7.67	31.44[b]	6.78	33.85[abc]	7.68	34.56[c]	6.97	34.60	7.59
ECW/ICW	–	0.84[a]	0.11	0.80[a]	0.12	0.81[a]	0.07	0.82[a]	0.07	0.82	0.10
OH (L)	–	0.62[a]	1.81	−0.44[b]	1.73	0.61[a]	1.36	0.17[c]	0.99	0.16	1.60
Dehydration% (n)	OH < −1.1 L	12 (10)[a]		39 (31)[b]		9 (2)[ac]		4 (3)[c]		17 (46)	
Hiperhydration % (n)	OH > + 1.1 L	30 (25)[a]		16 (13)[bc]		30 (7)[ab]		10 (8)[c]		20 (53)	

BCM, body cell mass; BISWB, spectroscopy multifrequency bioelectrical impedance by whole body analysis; CKD, chronic kidney disease; ECW, extracellular water; ICW, intracellular water; OH, overhydration; PhA, phase angle; Re, resistance; TBW, total body water; Xc, reactance.
Analysis of variance (quantitative variables) and χ^2 (qualitative variables) between CKD subgroups with adjusted P by Bonferroni test, $P \leq 0.008$ (values without a common letter [abc] differ significantly). Bioelectrical impedance data by BISWB
[1]PhA<6.11° for men and <5.14° for women [25].

Table 3
Body composition analysis of total sample and stratification by CKD subgroup

Variables	NDD (n = 83)		HD (n = 79)		PD (n = 23)		KTx (n = 81)		Total sample (N = 266)	
	X	SD	X	SD	X	SD	X	SD	X	SD
DXA										
ALMI (kg/m^2)	7.32^a	1.49	6.31^b	1.13	6.41^{ab}	1.50	6.84^a	1.29	6.79	1.39
LM (kg)	44.47^a	10.50	37.86^b	8.65	37.44^b	10.12	40.16^b	8.99	40.59	9.84
LMI (kg/m^2)	16.21^a	3.05	14.33^b	2.10	13.74^b	2.34	15.07^a	2.44	15.09	2.67
FFM (kg)	46.20^a	10.82	39.26^b	8.97	38.97^b	10.63	41.70^b	9.34	42.14	10.23
FFMI (kg/m^2)	16.82^a	3.14	14.84^b	2.15	14.28^b	2.43	15.63^b	2.52	15.65	2.76
FM (kg)	25.22^a	8.94	21.16^b	9.23	21.27^b	6.36	23.32^a	7.07	23.09	8.43
FMI (kg/m^2)	9.35^a	3.51	8.24^a	3.88	7.97^a	2.44	8.92^a	2.94	8.77	3.41
BISWB										
FFM (kg)	42.19^a	10.05	35.80^b	10.23	39.32^{ab}	9.80	39.02^{ab}	10.34	39.08	10.42
FFMI (kg/m^2)	15.35^a	2.83	13.45^b	2.84	14.45^{ab}	2.34	14.60^a	2.95	14.48	2.91
FM (kg)	25.27^a	11.18	19.12^b	6.84	23.20^{ab}	8.61	23.29^a	9.84	23.29	9.84
FMI (kg/m^2)	9.34^a	4.25	8.77 ^{a}	4.11	7.12^b	2.47	8.86^a	3.44	8.83	3.87

ALMI, appendicular lean mass index; BISWB, spectroscopy multifrequency bioelectrical impedance by whole body analysis; CKD, chronic kidney disease; DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass, FFMI, fat-free mass index; FM, fat mass; FMI, fat mass index; HD, hemodialysis; KTx, kidney transplant; LM, lean mass; LMI, lean mass index; NDD, non-dialysis–dependent; PD, peritoneal dialysis.
Analysis of variance between CKD subgroups with adjusted *P* by Bonferroni test, $P \leq 0.008$ (values without a common letter [abc] differ differ significantly)

For all CKD subgroups, the main cause of kidney disease was SAH. Fourteen patients (4 men and 10 women) were in automated and 9 (4 men and 5 women) in continuous ambulatory PD. KTx was by living donor in 18 patients (14 men and 4 women) and by deceased donor in 63 (34 men and 29 women). PD patients had less dialysis time than those on HD. Weight was higher in men than in women (77 ± 15 and 65 ± 13 kg, respectively; $P \leq 0.05$) and NDD had the highest weight of the CKD subgroups. WC was lower in women than in men (97 ± 12 and 100 ± 13 cm, respectively; $P \leq 0.05$).

BIS and DXA data of the total sample and CKD subgroups are shown in Tables 2 and 3. Low PhA was present in ~30% of the sample. NDD had the highest ECW and TBW and HD the lowest values, including OH. HD had the highest prevalence of dehydration and NDD and PD, of hyperhydration. Men had higher muscle mass and lower FM than women (ALMI: for men, 7.65 ± 1.13 and women, 5.88 ± 1.00 kg/m^2; $P \leq 0.05$; LM: for men, 47.15 ± 7.95 and women, 33.62 ± 6.21 kg; $P \leq 0.05$; LMI: for men, 16.47 ± 2.40, and women, 13.63 ± 2.12 kg/m^2; $P \leq 0.05$; FM: for men, 21.63 ± 8.36 and women, 24.65 ± 8.25 kg; $P \leq 0.05$; FMI: for men, 7.57 ± 2.94 and women, 10.05 ± 3.41 kg/m^2; $P \leq 0.05$). NDD had the highest FFM and LM; HD had the lowest ALMI, and FM was lower in patients on HD.

In the prospective analysis (10 ± 2 mo later than the first assessment), 87 patients were evaluated. Of the 87 evaluated, 25 (29%) were from the NDD group,18 (21%) from HD, 15 (17%) from PD, and 29 (35%) from KTx group. Of these patietns, 47 (54%) were men, with 16 (18%) in NDD, 6 (7%) in HD, 5 (6%) in PD, and 20 (23%) in the KTx groups. One patient from the NDD group started PD. One from HD underwent KTx. Seven patients from PD underwent KTx and two started HD. One patient from KTx started HD and the remaining 75 patients maintained the initial CKD treatment. For anthropometric and body composition changes (prospective – cross-sectional), the total demonstrated a weight loss of −0.91 ± 3.29 kg (NDD: −0.94 ± 3.03 kg; HD: −0.45 ± 2.92kg; PD: −1.78 ± 3.07k g, and KTx: −0.72 ± 3.87kg), a loss of FFM of −1.59±3.32, −1.54 ± 3.31, −2.19 ± 3.15, and −1.33 ± 3.66 kg for NDD, HD, PD, KTx, respectively. In all, the patients experienced a gain of FM of 0.59 ± 3.61 kg, with those in the NDD, HD, PD, and KTx groups seeing a gain of 0.39 ± 3.43, 0.88 ± 4.14, 0.46 ± 4., and 0.64 ± 3.37 kg. respectively. There were no differences in the CKD subgroups (ANOVA, $P > 0.008$) and sexes (unpaired *t* test $P > 0.05$).

Comparison of body composition measured by DXA and estimated by BIS

Table 4 shows the results of the agreement analysis between DXA and BISWB. BISWB underestimated FFM and slightly overestimated FM. Agreement between DXA and BISWB was higher for FM than FFM and better in cross-sectional data than in the evaluation of body composition changes. The Bland–Altman plot showed high LOA and a significant trend, which indicates increasing error with increase in muscle mass for FFM and high error in the low and high extremes of fat mass for FM; LOA were higher for FFM than FM (Fig. 1).

Supplementary Table 1 shows the agreement between DXA and BISSEG. BISSEG had worse agreement with DXA than BISWB. Supplementary Figure 1 shows Bland–Altman plots between DXA and BISSEG; high LOA and decreasing error were observed with increases in muscle mass for FFM and increasing error with increase in fat mass for FM.

We then investigated which variables were associated with bias between DXA and BISWB. The differences between the two methods were affected by ECW/ICW, FMI, BMI, WC, Re, and Xc for FFM (adjusted r^2 = 0.905) and FM (adjusted r^2 = 0.871), but with opposite association (Table 5). ECW/ICW was the variable that most strongly affected agreement, with higher values increasing FFM and decreasing FM bias. We stratified the sample into tertiles of ECW/ICW: In men, the first tertile (ECW/ICW ≤0.59 to <0.75) had an ECW/ICW ratio of 0.698 ± 0.040, the second tertile (ECW/ICW ≤0.75 to <0.83) an ECW/ICW ratio of 0.787 ± 0.024, and the last tertile (ECW/ICW ≥0.83) an ECW/ICW ratio of 0.917 ± 0.0829. In women, the first tertile (ECW/ICW ≤0.67 to <0.79) had an ECW/ICW ratio of 0.742 ± 0.032, the second tertile (ECW/ICW ≤0.79 to <0.88) an ECW/ICW ratio of 0.832 ± 0.028, and the last tertile (ECW/ICW ≥0.88) an ECW/ICW ratio of 0.954 ± 0.07. We then evaluated the agreement between methods among tertiles (Supplementary Table 2): Intensity and signal of the bias varied according to the ECW/ICW tertile (overestimation of FFM by BISWB in the first tertile and underestimation in the others; underestimation of FM in the first tertile and overestimation in the third). FFM agreement was much lower in the last tertile, and slight differences were observed for FM among tertiles. For body composition changes, we included only the patients who were within the same ECW/ICW tertiles in the prospective and initial evaluations.The

Table 4
Agreement between DXA and BISWB in total sample stratified by sex

Variables	Men															Women												
	DXA		BISWB		Bland–Altman analysis				ICC analysis			BISWB–DXA		5% Tolerance		DXA		BISWB		Bland–Altman analysis				ICC analysis			BISWB/DXA		5% Tolerance
					Bias[1]		LOA		ICC	95%CI										Bias[1]		LOA		ICC	95% CI				
	X	SD	X	SD	X	SD	Lower; upper		value	Lower; upper		X	SD	% (n)		X	SD	X	SD	X	SD	Lower; upper		value	Lower; upper		X	SD	% (n)
Cross-sectional data n = 137																Cross-sectional data n = 129													
FFM (kg)	48.99	8.23	46.68	7.67	2.31	7.22	−11.85; 16.47		0.566	0.425; 0.677		0.96[†]	0.14	23 (31)		34.87	6.42	31.01	5.91	3.86	5.32	−6.56; 14.29		0.525	0.179; 0.717		0.89[†]	0.14	22 (28)
FFMI (kg/m^2)	17.09	2.46	16.31	2.39	0.79	2.40	−4.09; 5.67		0.453	0.300; 0581		0.96[†]	0.14	23 (31)		14.11	2.18	12.54	2.03	1.58	2.17	−2.68; 5.84		0.365	0.075; 0.571		0.89[†]	0.14	22 (28)
FM (kg)	21.63	8.36	21.82	10.2	−0.19	4.31	−8.26; 8.64		0.894	0.855; 0.923		0.99	0.26	17 (24)[†]		24.65	8.25	24.85	9.21	−0.20	3.25	−6.58; 6.17		0.931	0.903; 0.950		1.00	0.18	29 (38)[†]
FMI (kg/m^2)	7.57	2.94	7.62	3.55	−0.06	1.51	−3.02; 2.90		0.894	0.854; 0.923		0.99	0.26	17 (24)[†]		10.05	3.41	10.12	3.78	−0.09	1.34	−2.72; 2.54		0.931	0.903; 0.950		1.00	0.18	29 (38)[†]
Δ data n = 47																Δ data n = 40													
FFM (kg)	−1.48	3.19	−0.40	4.09	−1.09	4.57	−10.04; 7.85		0.196	−0.109; 0.471		0.50	3.42	0 (0)		−1.71	3.50	−0.67	3.02	−1.17	3.23	−7.50; 5.15		0.495	0.214; 0.602		0.70	2.06	3 (1)
FFMI (kg/m^2)	−0.46	1.15	−0.12	1.46	−0.35	1.63	−3.54; 2.84		0.233	−0.073; 0.501		0.22	3.00	3 (1)		−0.65	1.44	−0.29	1.22	−0.45	1.28	−2.95; 2.05		0.528	0.246; 0.729		0.18	2.45	0 (0)
FM (kg)	0.82	4.02	0.36	4.44	0.36	4.37	−8.20; 8.93		0.465	0.181; 0.663		0.77	2.75	2 (1)		0.33	3.12	−0.44	3.37	0.82	2.94	−4.95; 6.59		0.582	0.326; 0.759		0.63	2.59	0 (0)
FMI (kg/m^2)	0.34	1.42	0.23	1.48	0.08	1.46	−2.78; 2.94		0.471	0.196; 0.677		0.38	2.68	5 (2)		0.15	1.35	−0.26	1.38	0.42	1.19	−1.91; 2.75		0.590	0.343; 0.760		0.79	2.82	0 (0)

BISWB, spectroscopy multifrequency bioelectrical impedance by whole body analysis; Δdata, body composition changes evaluated as prospective − cross-sectional measurements; DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass, FFMI, fat-free mass index; FM, fat mass; FMI, fat mass index; ICC, intraclass correlation coefficient between DXA and BISWB measurements with 95% CI; LOA, limits of agreement by Bland–Altman analysis (lower limit; upper limit).
[1] Bias calculates as DXA–BISWB measurements; 5% tolerance between DXA and BISWB measurements calculate as BISWB/DXA ≤0.95 to ≤1.05.
[†] unpaired Student t-test (quantitative variables) and χ^2 (qualitative variables) between sexes, $P \leq 0.05$.

first tertile for men had better ICC for FFM and FM data. For women, the second tertile had better ICC for FFM and the last tertile a slightly better ICC for FM.

Prediction of DXA values of body composition by BISWB method

As ECW/ICW was the most influential factor of the agreement between DXA and BISWB (highest coefficient), we stratified the total sample into design and validation samples to identify the cut-off point for ECW/ICW above which the agreement between the methods would have an inadequate error tolerance (DXA/BISWB > ±2 kg) and the use of BISWB for body composition evaluation would not be indicated. This analysis was done only for FFM because FM obtained high ICC and relatively low LOA.

Design and validation samples showed no differences in clinical, BIS, and DXA characteristics (Supplementary Table 3). The area under the ROC curve for ECW/ICW (Supplementary Fig. 2) was 0.65 with a cutoff point of ECW/ICW ≥0.7250 for inadequate error tolerance between DXA and BISWB for FFM data, which was validated in the validation sample having a sensitivity of 80 (95% CI, 66−93) and a specificity of 30 (95% CI, 9−51).

To best predict the DXA estimation of body composition by BISWB measurement, we fitted multiple linear regression models and chose the one with the best prediction while maintaining a small number of variables that are easily obtained (Table 6). Because FFM was not normally distributed in the design sample, a log transformation was applied. The final models derived from the design sample were as folow:

$$\begin{aligned}\mathrm{FFM(kg)} = \mathrm{e}(&1.26010 + 0.08458 \times (\mathrm{sex}) + 0.00019 \times (\mathrm{age}) \\ &+ 0.01131 \times (\mathrm{BMI}) + 0.00307 \times (\mathrm{WC}) + 0.00023 \\ &\times (\mathrm{Re\,50\,kHz}) + 0.00242 \times (\mathrm{Xc\,50\,kHz}) + 0.90554 \\ &\times (\mathrm{ECW/ICW}) + 0.02039 \times (\mathrm{FFM}))\end{aligned}$$

$$\begin{aligned}\mathrm{FM(kg)} = 31.79970 &- 2.92130 \times (\mathrm{sex}) - 0.02109 \times (\mathrm{age}) - 0.18604 \\ &\times (\mathrm{BMI}) + 0.04547 \times (\mathrm{WC}) - 0.01221 \times (\mathrm{Re\,50\,kHz}) \\ &- 0.06439 \times (\mathrm{Xc\,50\,kHz}) - 19.17652 \times (\mathrm{ECW/ICW}) \\ &+ 0.86347 \times (\mathrm{FM})\end{aligned}$$

where sex is 1 for men and 0 for women; age in years; BMI in kg/m^2; WC in cm; Re and Xc in ohm; FM and FFM in kg obtained by original BISWB prediction.

Before and after applying the new models, the differences between the two methods were compared based on the percentage of the sample for which the residual (DXA − BISWB data) was <2 kg (Fig. 2). After applying the new model, for FFM, the percentage of residuals <2 kg increased from 30% to 55%, and for FM, it increased from 39% to 63%.

Discussion

In the present study, a higher agreement between BIS and DXA in a multi-treatment CKD sample was found when using BISWB compared with BISSEG analysis, for FM compared with FFM data, and when using cross-sectional data compared with body compositions changes. Interfering factors were ECW/ICW ratio, FMI, BMI, WC, Re, and Xc in descending order of importance. Although sex was not a significant interfering factor, LOA was lower among women than men for FM and FFM. ECW/ICW ratio was the most

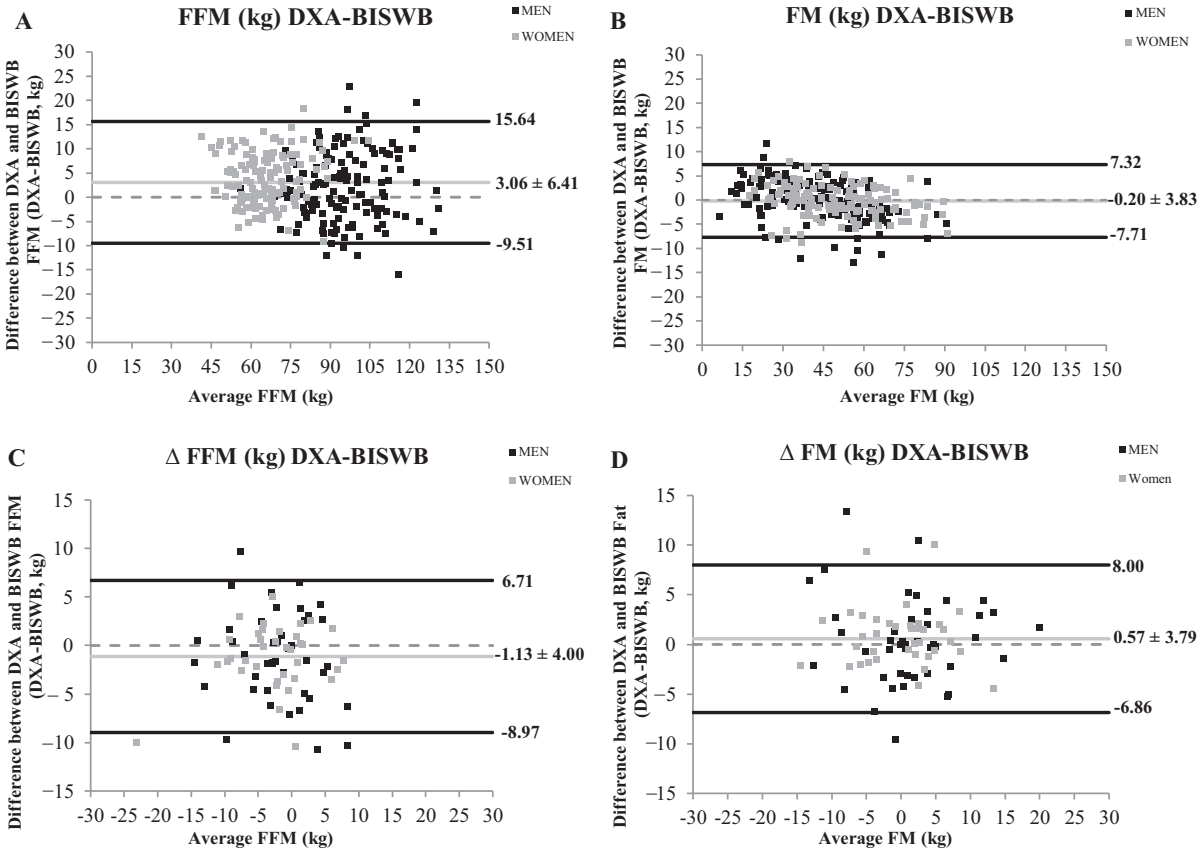

Fig. 1. Bland–Altman plot with sex stratification for comparison between DXA and BISWB. (**A**) FFM data by cross-sectional measurements. (**B**) FM data by cross-sectional measurements. (**C**) FFM data by prospective – cross-sectional measurements (Δ). (**D**) FM data by prospective – cross-sectional measurements (Δ). Solid black horizontal lines are limits of agreement; solid grey horizontal line is the median of bias between methods. BISWB, spectroscopy multifrequency bioelectrical impedance by whole body analysis; DXA, dual energy x-ray absorptiometry; FFM, fat-free mass; FM, fat mass.

influential factor for bias between BISWB and DXA with worse agreement in FFM for the last tertile of ECW/ICW-stratified sample. Thus, a cutoff point for ECW/ICW ≥0.7250 was established for inadequate error tolerance among methods. New equations for FFM and FM were developed and validated with better performance than the original ones.

Although the results varied, differences between bioimpedance and DXA body composition data were observed both in patients with normal and altered kidney function. Similarly, Lee and et al. [13] found high LOA for muscle mass data (−5.55 to 8.10 kg) and increasing error with increases in muscle mass when assessing agreement between bioimpedance and DXA assessments. Additionally, differences between methods were affected by BMI and body fat. As FFM data obtained by DXA (47 ±6.5 and 34 ± 4.8 kg for men and women, respectively) were similar to our sample (48.99 ± 8.23 and 34.87 ± 6.62 kg, for men and women respectively). The lower LOA value in the Lee et al. study was probably due to an absence of hydration disturbance in their sample (estimated glomerular filtration rate ~90 mL/min/1.73 m^2, with the exclusion of HD and PD patients). These results show the importance of fluid overload in worsening the agreement between measures, and FM (although also affecting bias in our study) as a major factor in populations without hydration disturbance (as in the Lee et al. sample).

Agreement between methods across the range of BMI was also analyzed by Day et al. [12], who found better agreement for FM data even in a sample of healthy adults. In their study, overweight and obese showed narrower LOA for FM than the normal weight group, whereas in our study, agreement was worst in the extremes of body fat. The difference between findings could be explained by the worse ECW/ICW ratio in our overweight and obese samples (0.823 ± 0.11, 0.840±0.09 for men and 0.847 ± 0.10, 0.871 ± 0.07 for women) compared with normal weight individuals (0.752±0.08 for men and 0.833±0.11 for women), and thus ECW/ICW overlapped with FM as the most important factor for bias between devices.

In a study with >3000 patients and a wide BMI range, normal weight individuals had lower bias and LOA than overweight and obese patients [14], as was similar to our data. However, in that study, FFM was overestimated and FM underestimated by bioimpedance in overweight and obese individuals, contrary to our results. As shown by our regression analysis, a higher ECW/ICW ratio leads to a higher DXA–BISWB difference for FFM (BISWB underestimates the values) and a lower for FM (BISWB overestimates the values). As overweight and obese had a higher ECW/ICW ratio than normal weight, limitations of BISWB in overweight and obese CKD patients may be explained by hydration variability.

A previous study compared BIS (BCM, Fresenius) with DXA for evaluation of body composition in NDD patients with higher ECW/ICW ratio (0.9 ± 0.1) and OH (1.2 ± 1 L) than our NDD sample [15]. The authors reported bias between data for FFM and FM with a positive and negative relationship with ECW/ICW, respectively, in accordance with our findings. Moreover, a 0.1 increase of ECW/ICW was associated with a 2.22 kg increase in FFM and 2.43 kg decrease in bias for FM, similar but lower than our findings (+5.03 kg for FFM and −2.79 kg for FM) probably because our results are based on the whole sample (several CKD treatment groups) having a much greater variation of fluid load.

Table 5
Simple and multiple linear regression models for factors associated with bias in agreement between DXA and BISWB

	FFM kg (DXA – BISWB)								FM kg (DXA – BISWB)							
	Simple linear regression model				Multiple linear regression model				Simple linear regression model				Multiple linear regression model			
	Coefficient (β)	95% CI		*P*-value	Coefficient (β)	95% CI		*P*-value	Coefficient (β)	95% CI		*P*-value	Coefficient (β)	95% CI		*P*-value
		Lower	Upper			Lower	Upper			Lower	Upper			Lower	Upper	
Intercept	–	–	–	–	−117.064	−124.339	−109.795	0.000	–	–	–	–	85.084	80.005	90.162	0.000
BMI (kg/m^2)	0.755	0.628	0.881	0.000	1.817	1.627	2.007	0.000	−0.243	−0.332	−0.155	0.000	−1.723	−1.855	−1.590	0.000
WC (cm)	0.322	0.274	0.371	0.000	0.168	0.122	0.213	0.000	−0.113	−0.148	−0.078	0.000	−0.079	−0.111	−0.047	0.000
Re50kHz (ohm)	−0.007	−0.014	0.000	0.064	0.049	0.041	0.056	0.000	−0.002	−0.006	0.003	0.053	−0.046	−0.051	−0.040	0.000
Xc 50kHz (ohm)	−0.206	−0.261	−0.152	0.000	0.091	0.016	0.166	0.017	0.053	0.018	0.089	0.003	−0.064	−0.117	−0.012	0.016
ECW/ICW	42.660	37.340	47.979	0.000	50.353	43.182	57.524	0.000	−16.129	−20.106	−12.152	0.000	−27.973	−32.982	−22.963	0.000
FMI (kg/m^2)	0.960	0.764	1.157	0.000	−1.951	−2.141	−1.762	0.000	−0.164	−0.299	−0.029	0.017	2.195	2.062	2.327	0.000

BISWB, spectroscopy multifrequency bioelectrical impedance by whole body analysis; BMI, body mass index; DXA, dual energy x-ray absorptiometry; ECW/ICW, extracellular to intracellular water ratio; FFM, fat-free mass; FM, fat mass; FMI, fat mass index (data by DXA analysis); Re, resistance; Xc, reactance; WC, waist circumference.
For regression model analysis, the following variables were evaluated: sex (1 man, 0 woman); ethnicity (1 white, 0 non-white); age (years); Diabetes Mellitus (1 existing, 0 nonexistent); systemic arterial hypertension (1 existing, 0 nonexistent); dyslipidemia (1 existing, 0 nonexistent); estimated glomerular filtration ratio (mL/min/1.73m^2); KT/V; creatinine (mg/dL); urea (mg/dL); weight (kg), height (m); overhydration index (L); extracellular water (L); intracellular water (L); total body water (L); extracellular to total body water ratio; body cell mass (kg); lean mass, fat-free mass and fat mass by DXA analysis as well as their indexes; the variables present in this table

Fluid overload and water disturbance is common in advanced CKD. Other studies [15], as well as our own, showed that measurement differences were associated with the ECW/ICW ratio. As NDD was the group with the higher prevalence of OH, followed by PD, BIS analysis of body composition could show higher bias in these patients. Fluid overload, even in the absence of inflammation, is a significant risk factor for short-term mortality [29]. Therefore, the association of body composition with mortality in CKD could be biased depending on the method applied.

A study developed with PD patients that also compared BIS (BCM, Fresenius) with DXA data [16], observed increasing bias with higher muscle mass and large individual variability among methods resulting in lower agreement on individual level, similar to our results. Compared with our findings for the PD subsample, the Popovic study found similar bias for FFM (−0.3 ± 3.8 versus −0.3 ± 5.6 kg) and a lower bias for FM (2.1 ± 2.7 versus 0.9±5.7 kg), respectively. However, our LOAs (from −7 to 8; and from −2 to 7, for FFM and FM, respectively) were lower. Differences among studies may be due to the different clinical characteristics of samples, as Popovic's patients were older (63 ± 13.5 y), had higher body weight (81 ± 18 kg), BMI (27 ± 5.5 kg/m^2), OH (1.6 ± 2.1 L), FM (26.9 ± 12.4 kg), and FFM (53.4 ± 11. 3kg) by BIS and lower Kt/V (2 ± 0.5) than our PD subgroup.

In a sample of older HD and NDD patients [17], BIS (BCM, Fresenius) overestimated FM compared with DXA, and age was the strongest factor to influence the variability of measurements, probably because that study evaluated a higher age range and measurements were done before HD sessions (OH = 1.3 ± 1.6 L). In our analysis, with measurements done after mid-week HD sessions, BIS underestimated FM. Differences between studies are probably related to the measurement protocols, which affected water disturbance and body composition prediction [30].

We were unable to locate studies that compared BIS and DXA for evaluation of changes in body composition of patients with CKD. Nilanjana Tewari et al. [31] evaluated agreement between single-frequency bioimpedance and DXA in 47 patients submitted to major abdominal surgery or neoadjuvant chemotherapy for esophagogastric cancer. Although applying different equipment and prediction equations in another disease population sample, the authors found results similar to ours with high bias (−3.79 kg) and LOA (from −15.52 to 7.93) for FFM data between methods for evaluation of body composition changes.

New prediction equations were developed for FFM and FM derived by BIS data and validated against DXA. The factors that interfered in the bias between methods were included in the new equations, as correction variables of the original prediction. As they showed better performance, we strongly encourage the implementation of these new equations in body composition assessment of adults with CKD when the BCM equipment is applied, especially in patients with ECW/ICW ratio ≥0.725, when inadequate error tolerance was observed. However, as we were not able to achieve adequate tolerance as found by Lee et al. [13], probably because our sample was a disease population with high rates of water imbalance, the equations should be used with caution in patients with fluid overload.

The basic assumption that the human body is a single, symmetrical cylinder with homogeneous composition and uniform cross-sectional area is far from being physiologically accurate. The body can best be described as being composed of five distinct cylinders with different resistivity. Moreover, as extremities contribute to about 90% of BISWB resistance measurement and the trunk to the remaining 10%, disproportionate to body mass distribution, we presumed that segmental analysis would provide a more accurate evaluation of body composition. Given that BISSEG performance

Table 6
Multiple linear regression models for prediction of FFM and FM by BISWB

	lnFFM (kg)					FM (kg)			
	Coefficient (β)	95% CI		*P*-value		Coefficient (β)	95% CI		*P*-value
		Lower	Upper				Lower	Upper	
Intercept	1.26010	0.822	1.698	0.000	Intercept	31.79970	12.576	51.023	0.001
Sex (men: 1; women: 0)	0.08458	0.553	0.116	0.000	Sex (men: 1; women: 0)	−2.92130	−4.105	−1.737	0.000
Age (y)	0.00019	−0.001	0.001	0.737	Age (y)	−0.02109	−0.066	0.024	0.356
BMI (kg/m^2)	0.01131	0.066	0.017	0.000	BMI (kg/m^2)	−0.18604	−0.524	0.152	0.278
WC (cm)	0.00307	0.001	0.005	0.004	WC (cm)	0.04547	−0.049	0.140	0.342
Re 50kHz (ohm)	0.00023	0.000	0.001	0.213	Re 50kHz (ohm)	−0.01221	−0.027	0.002	0.095
Xc 50kHz (ohm)	0.00242	−0.001	0.006	0.128	Xc 50kHz (ohm)	−0.06439	−0.191	0.062	0.316
ECW/ICW	0.90554	0.585	1.226	0.000	ECW/ICW	−19.17652	−31.768	−6.585	0.003
FFM (kg)[1]	0.02039	0.017	0.023	0.000	FM (kg)[1]	0.86347	0.690	1.037	0.000
Adjusted r^2	0.913				0.887				

BMI, body mass index; BISWB, spectroscopy multifrequency bioelectrical impedance by whole body analysis; ECW/ICW, extracellular to intracellular water ratio; FFM, fat-free mass; FM, fat mass; Re, resistance; Xc, reactance; WC, waist circumference
[1]FFM and FM (kg) data by BISWB.

was worse than that of BISWB, we speculated whether the segment cutoffs between BISSEG and DXA were equal. Additionally, the BISSEG analysis assumes that both arms and both legs are equal; whereas DXA measures the extremities individually (each arm and each leg)

As BMI could reflect fluid overload, we preferred to use WC as an obesity index, which is associated to increased cardiovascular risk. WC was high in 49% (NDD: 60%; HD: 29%; PD: 37%; and KTx: 54%) and 76% (NDD: 68%; HD: 82%; PD: 60%; and KTx: 85%) of male and female samples, respectively. This data is in agreement with

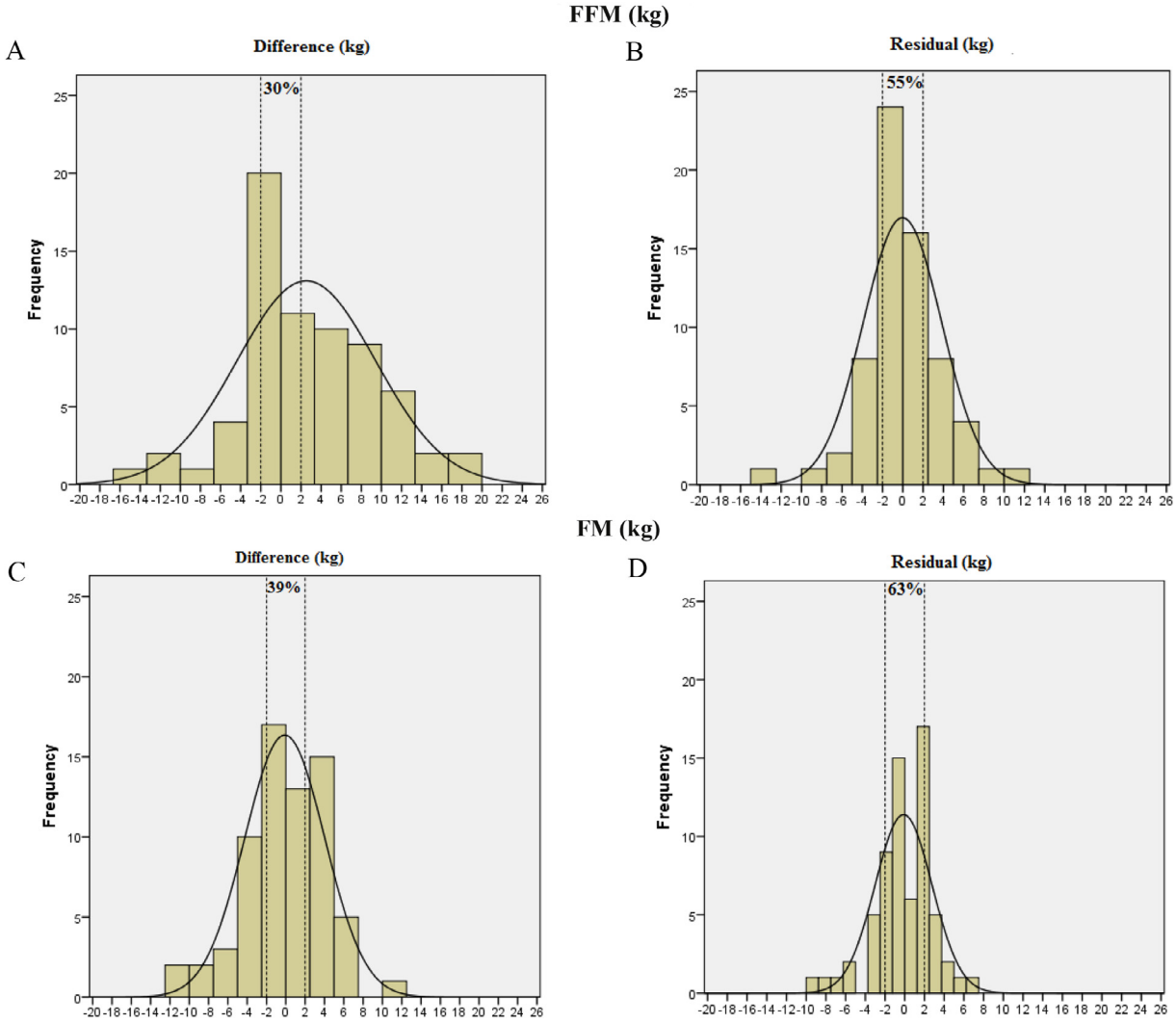

Fig. 2. Comparing the percentages of residuals <2 kg between expectations before (**A, C**) and after (**B, D**) applying the new regression formulae in the validation sample. See Table 6 for the new prediction equation for FFM and FM. See Supplementary Table 7 for data about the design and validation samples. FFM, fat-free mass; FM fat mass.

the higher prevalence of obesity in end-stage CKD reported in the last 20 y [32]. Obesity could be a protective factor in patients on dialysis with dependency of the presence of adequate muscle mass [33], and a harmful one in NDD and KTx patients [8]. Our HD and PD samples had lower levels of FM compared with the NDD and KTx patients, who had the highest levels. Unfortunately, in the prospective analysis, all groups lost FFM and gained FM, showing a worsening of nutritional status in CKD independent of the treatment applied.

Also, low muscle mass and sarcopenia have higher prevalence in CKD [6] compared with the general population [34]. Our dialysis patients, especially those in HD, had the lowest muscle mass (ALMI, LM, FFM, and FFMI by DXA), in accordance with previous studies showing higher prevalence of low muscle mass and sarcopenia among dialysis patients with an increase of muscle depletion as kidney function decreases [35]. Taking into account body composition analysis, HD patients presented the worst nutritional status, albeit PD analysis may have been jeopardized by the low number of patients. With a predialysis nutritional depletion [3], patients are further affected by the side effects of HD therapy that include loss of amino acids and proteins in the dialysate added to a higher inflammatory condition and increased metabolic acidosis [36], partially explaining the worse body composition of these patients.

Although many studies have previously compared measurements of body composition by DXA and bioimpedance, to our knowledge, this is the only one comparing BIS and DXA results in CKD patients in the four treatment groups of NDD, HD, PD, and KTx, allowing a wide range of hydration states, providing better understanding of the interfering factors in the bias between methods. We applied BIS as bioimpedance method, which is more reliable for body composition assessment than single-frequency equipment [10], as BIS can distinguish between ECW and ICW within a three-body compartment model (FM, FFM, and water) whereas the single-frequency machines apply the two-body compartmental model [37], being BIS less affected by hydration status [37]. Moreover, we evaluated whole body and segmental BIS, FFM and FM tissues, and collected cross-sectional and prospective data to assess body composition changes, allowing a comprehensive and in-depth analyses of BIS performance. DXA, a reference method for body composition analysis, was applied in all participants. Thorough statistics were carried out permitting a better understanding of the appropriate use of BIS body composition evaluation and settings: agreement analysis, graphical concordance analysis, evaluation of factors affecting bias between methods, and the definition of a cutoff point for inadequate error tolerance. The newly developed equations have a better performance and might help clinicians estimate muscle and FM more accurately.

However, there were several limitations to this study. We did not include elderly individuals, so the results and the new equations can only be extrapolated to adults with CKD. Although we had a large sample size, the PD group was underrepresented, which may have affected CKD subgroup comparisons. The prospective analysis had fewer participants than the cross-sectional one and was done only once; investigations with larger sample sizes and frequencies and of longer durations in longitudinal assessments are needed. As estimation of body composition by bioelectrical impedance is based on frequency and mathematical models specific for certain equipment [38], the new equations and observed performances should be used for BIS measured in the BCM Fresenius Medical Care machine. Although widely considered a reference method for assessing body composition, DXA is also affected by water disturbances, although in a much lower degree than bioelectrical impedance. Because the ratio of additional body fluid is similar to that of LM, DXA has been shown to overestimate LM in high fluid overload status [39]. Lastly, we improved the prediction of FFM and FM by BIS; however, a significant percent of the sample was outside the adequate error tolerance range, thus, cross-validation should still be performed.

Conclusion

Bioelectrical impedance techniques are non-invasive, inexpensive, and simple to use; however, they apply various equations and assumptions for body composition estimation. Overall agreement was good on a population basis, mainly for FM data, but agreement was poor on an individual level. For body composition estimates in CKD, BIS should be applied in the whole body protocol, and after the HD session and drainage of peritoneal fluid to decrease bias, as ECW/ICW was the most influent factor in the agreement with DXA. In normal hydration CKD patients, BIS is as reliable as DXA; however, BIS has to be used with caution in OH patients, especially for FFM data. An exploratory cutoff point of ECW/ICW $\geq$0.7250 was established but needs to be validated. Longitudinal assessment showed a poorer agreement with DXA compared with the cross-sectional evaluation. The new formulas improved prediction, although further validation is needed; these equations might offer a better option for body composition estimation, primarily FFM.

Moreover, as methods for body composition analysis are not interchangeable in patients with CKD, we suggest using the same measure of body composition over time in a standardized protocol and application time as a way to minimize bias.

Declaration of Interests

Accountability's Work Group on Medical Students in the Class of 2021 Moving Across Institutions for Post Graduate Train The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.nut.2020.111059.

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